

Lukas Paul Robertson

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Personal Profile

Genetics student at UAB (top 5% of cohort) with parallel training in informatics. Strong programmer with research experience in deep learning, computational biology, and literature reviews. Entrepreneurial experience in non-profits. Seeking opportunities to work in technical digital minds research. Generally interested in all things Bio $\cup$ Neuro $\cup$ CS.

Education

- 2023 – 2027 **B.Sc. Genetics, Universitat Autònoma de Barcelona (UAB).**
GPA: 8.56/10 (Top 5% of cohort) as an international student in a Catalan-taught program. Highly selective with the highest state-wide entry requirements for a biosciences degree.
- 2024 – 2027 **Informatics, 42 Barcelona.**
Programming school based on peer-to-peer feedback and autodidactic learning.

Research Experience

- Jul 2026 – present **Scientific Trainee**, EMBL Heidelberg, Stegle Lab.
 - Led an independent project: reviewed the literature, identified a gap, and developed a novel transformer architecture for 3D medical imaging.
 - Built and maintained the training pipeline on EMBL's GPU cluster.
 - Coordinated a collaboration with researchers at Uniklinikum Erlangen.
 - Presented preliminary results to the full lab; preparing a manuscript.
- Jun 2025 – Sep 2025 **Intern**, IBMB, Perez Lab.
 - Joined a lab with no prior ML experience; tasked to use ML to investigate *C. elegans* gene regulation
 - Independently designed and created a flexible ML pipeline with data cleaning, feature engineering, and parallelized GPU acceleration
 - Established the lab's ML capabilities; lab continues to pursue this research direction
- Aug 2024 – Apr 2025 **Founding Researcher**, Screwworm Free Future.
 - Investigated technical bottlenecks and tractability of *Cochliomyia hominivorax* eradication
 - Conducted literature review, consulted with researchers, and synthesised findings
 - Findings included in slide-decks and communications that helped raise \$150,000
 - Contributed to broader entrepreneurial tasks including communications, operations, and hiring.

Technical Skills

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|---------------|--|
| Programming | Python, C/C++, R, Bash, scripting, $\LaTeX$, Docker. |
| Deep learning | Transformers, LLMs, CNNs, parallel/GPU computing, Pytorch, Numpy. |
| ML/Data | Optuna, XGBoost, Dask, Pandas, SLURM. |
| Biology | Bioinformatics, cloning/genetic engineering, PCR, gel electrophoresis, bacterial cultures. |

Other Experience

Mar 2024 – Mar 2025	Founder & President , Altruisme Efectiu UAB. • Founded and led the student organization • Designed and ran a 7-week fellowship that gave participants 2 ECTS • Organised speaker events and mentored newer members
Jun 2024 – Present	Organizer , AI Safety Barcelona. • Founding member of AI Safety Barcelona • Help organise events, outreach, and headhunting
Jul 2023 – Jan 2024	Translator/Transcriber . • Translated and transcribed interviews in English, Spanish, and French • Worked between stakeholders and project beneficiaries for a humanitarian aid consultant

Languages & Courses

Languages	English (Native) • Spanish (Native) • Catalan (Professional) • Dutch (Limited) • French (Limited)
ML/Data	XGBoost (Datacamp) • Parallel Programming with Dask (Datacamp) • Hyperparameter Optimization with Optuna (Datacamp) • Algorithms and Data Structures (FreeCodeCamp)
Other	Principles of Economics: Micro & Macro (MRU) • Biosecurity Fundamentals (Blue Dot Impact)

Other Activities

Dec 2025 – Present	Representative of Biosciencias UAB. Represent ~2000 bioscience students in the university-wide forum (<i>Claustre UAB</i>) as an independent without party backing. Also represent ~300 Genetics students in the bioscience faculty-wide forum.
Sep 2024 – Present	Population Genetics Mentorship. One of two students receiving mentorship from Prof. Antonio Barbadilla, an opportunity usually given to final year students. Offered on the basis of academic achievement.
Feb 2025 – Present	Advisor, Karoteno. Advise the Spanish effective giving initiative Karoteno on strategy, tech, and outreach. Recruited for previous experience in non-profit entrepreneurship.
Feb 2024	Global Challenge Priorities Attendant. Received \$200 stipend to attend GCP workshop in Oxford, England.

References

- ☑ **Prof. Antonio Barbadilla** — Bioinformàtica de la Diversitat Genòmica, IBB, UAB
- ☑ **Prof. José Domínguez** — One and a Half Group, IGTP, UAB
- ☑ **Melanie Brennan** — EA Barcelona Organizer